

INLP Course Project

Evaluating machine unlearning under
quantisation and adversarial probing

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Problem Statement

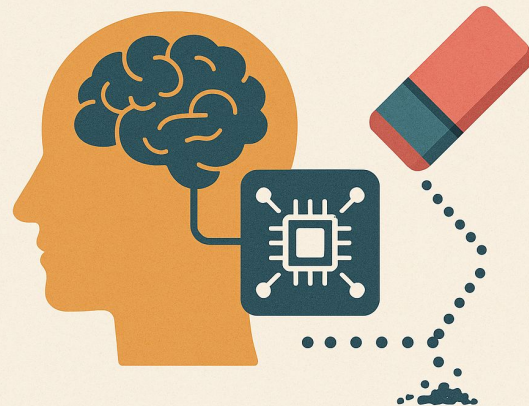
Research Objective:

Investigating the robustness of machine unlearning under low-bit quantization and adversarial probing.

Methodology & Setup

- Model: Gemma-3-1B architecture
- Benchmark: MUSE (Harry Potter corpus as proxy)
- Unlearning methods
 - Gradient ascent with NPO
 - Task vector

MACHINE UNLEARNING



Gemma 3



Why we picked this?



- LLMs train on sensitive / copy wrieded data
- LLM are not deployed without quantisation in applications
- LLMs can't be retrained from scratch on copyright strikes
 - Unlearning vs Retraining

Unlearning and Methods



What is good unlearning? (MUSE paper)

Core Idea:

Unlearning makes the model unable to recall any data from the “forget set” and “retain” all the other data

Goals:

- **No Verbatim Memorization:** Must not be able to reproduce any passages from the “forget set”.
- **No Knowledge Memorization:** Be completely incapable of correctly answering questions based on the specific knowledge
- **No Privacy Leakage:** Must not leave traces that could be exploited. (Adversarial probing)
- **Utility Preservation:** General capabilities and performance on the “retain set” shouldn’t be affected
- **Scalability:** Must be capable of handling large-scale content removal effectively
- **Sustainability:** Must be able to handle successive, sequential unlearning requests over time without loss in performance.



We focused on the first 4 criteria for our project.

Gradient Unlearning: Negative Preference Optimization (NPO)

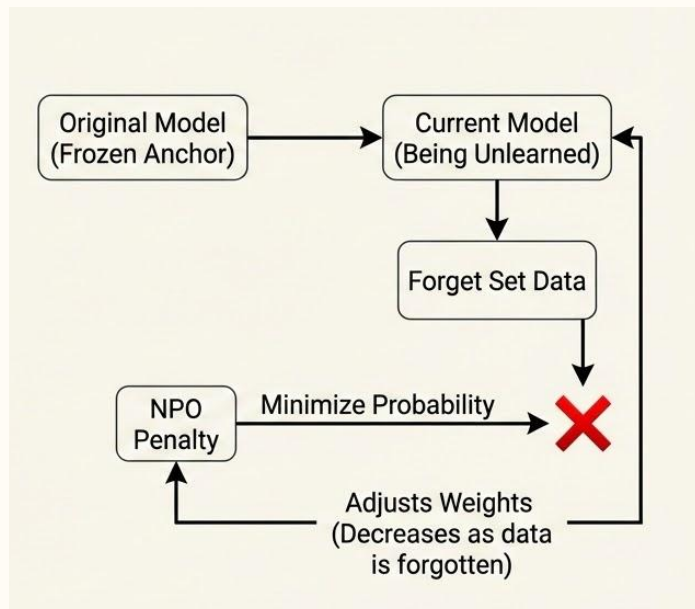
Core Idea: Built on Gradient ascent but without its instability of Simple GA

Mechanism:

- Treats exact passages from the "forget set" as rejected responses.
- Opposite of direct preference optimisation

How NPO (Negative Preference Optimization) Works

- **The Mechanism:** NPO tries to solve the catastrophic collapse caused by GA. It treats the data to be forgotten as a "negative preference" (an output the model should avoid).
- **The Anchor:** Instead of blindly maximizing error, NPO introduces a frozen copy of the original model (the "reference model").
- **The Speed Limit:** When unlearning a piece of data, NPO compares its current output probability to the reference model's probability. As the model successfully "forgets" the target text, the mathematical penalty naturally decreases to zero.



Other optimisations:

- Gradual Unfreezing
- C-NPO
- β parameter tuning
- Adaptive LR on the unlearning

Geometric Unlearning: Task Vector Arithmetic

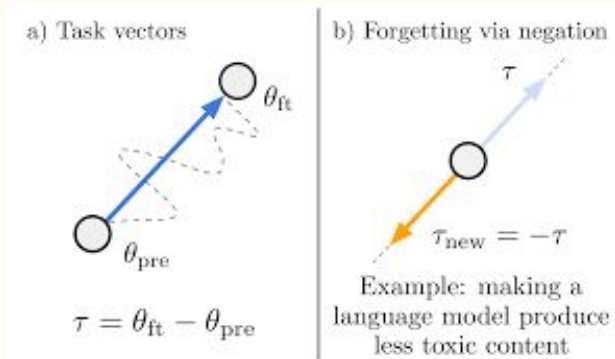
Core Idea: Discards complex loss landscapes in favor of leveraging the geometric properties of the model's weight space.

Geometric Approach:

- **Weight Space as a Map:** Views fine-tuning as a directed vector within the model's high-dimensional parameter space.
- **Isolating "Knowledge":** Extracts a specific direction that corresponds to the model's memorized "forget set."

How to achieve it:

- **Fine-Tuning:** Create an intentionally overfitted model (θ_{ft}) on the forget set.
- **Vector Extraction:** Compute the Task Vector: $\tau = \theta_{ft} - \theta_{pre}$. This vector isolates the specific parameters responsible for the target data.
- **Geometric Negation:** Subtract the scaled task vector from the base weights: $\theta_{unlearn} = \theta_{pre} - \lambda\tau$



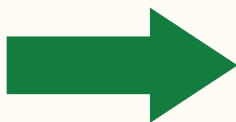
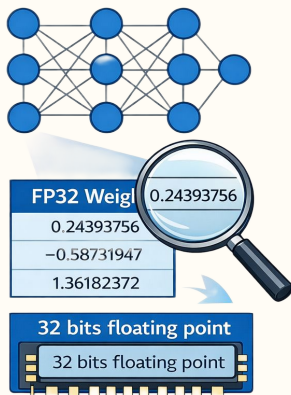
Quantization



Quantization

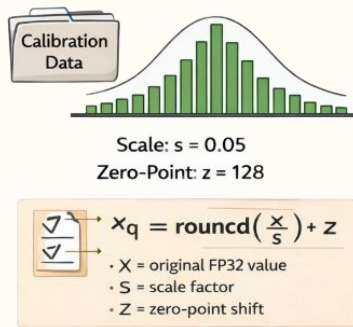
What is Quantization?

Quantization is the process of reducing the numerical Precision of the Neural Network Parameters, for example conversion from FP16 to INT8/4 bit representation.



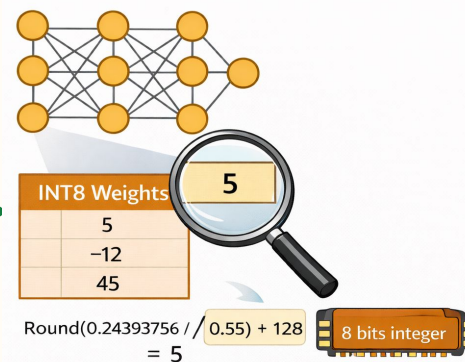
Why is Quantization?

- Reduces Model Size.
- Faster Inferences.
- Easier Deployment to Edge Devices.
- Lower energy and RAM requirements,



Who is Quantization?

Quantization comes in various forms, the one we are using is Post Training Quantization, where after training is done we quantize the model. Both the weights and activation values are recalibrated and then the model is deployed.



3. Quantize to INT8

Effects of Post Training Quantization

Weight Update Collapse

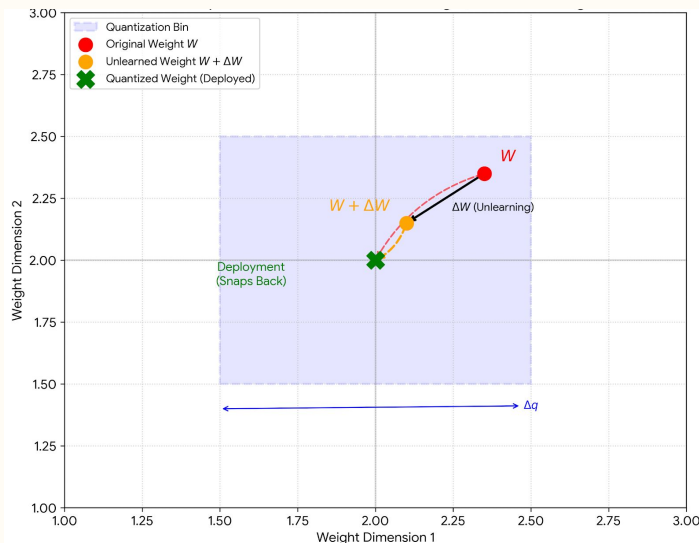
Unlearning changes the weights by a small updates, and quantization tries to fit these continuous weights into discrete levels. If the updates are too small the original wt and the changed wt may get quantized to the same level.

Knowledge Re-emergence

Because the unlearning steps updates are inherently small, post quantization the model may "snap back to the same weight as before" making the knowledge retained, instead of forgotten.

Metrics used by us

We use Recovery Delta metric in our work. It is the mathematical difference in forgetting metric (like ROUGE-L) between the quantized unlearned and the unquantized unlearned models.



Types of Quantization Implemented

FP16 to NF8

Method: Utilizes the `bitsandbytes` library to compress 16-bit float weights down to 8-bit.

Compute Strategy: Weights are stored in 8-bit, but dequantized to FP16 (`bnb_8bit_compute_dtype=torch.float16`) during forward passes to maintain generation quality.

Memory Impact: Halves the VRAM footprint of the model.

Evaluation Goal: To see if with mild compression can the data remain forgotten or it reemerges?

FP16 to NF4

Method: Uses 4-bit `NormalFloat` (NF4), an information-theoretically optimal data type for normally distributed model weights.

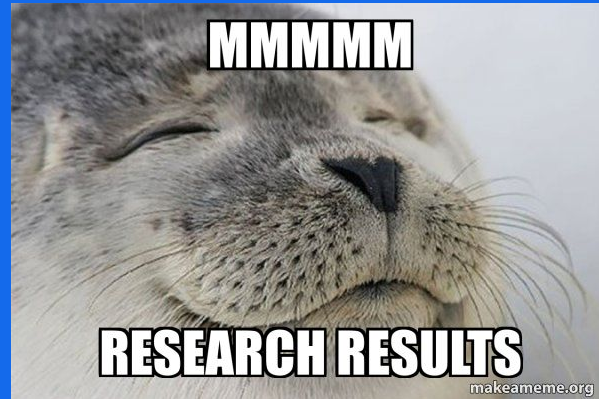
Double Quantization: Implements nested quantization (`bnb_4bit_use_double_quant=True`) to further compress the quantization constants themselves.

Memory Impact: Reduces VRAM requirements by ~75%, allowing massive models to run on consumer GPUs.

Evaluation Goal: To See if the Model breaks by such compression and to check if the knowledge reemerges. (Due to more reduction, chances of snapping back can be increased).

Our Work

What we did + Results



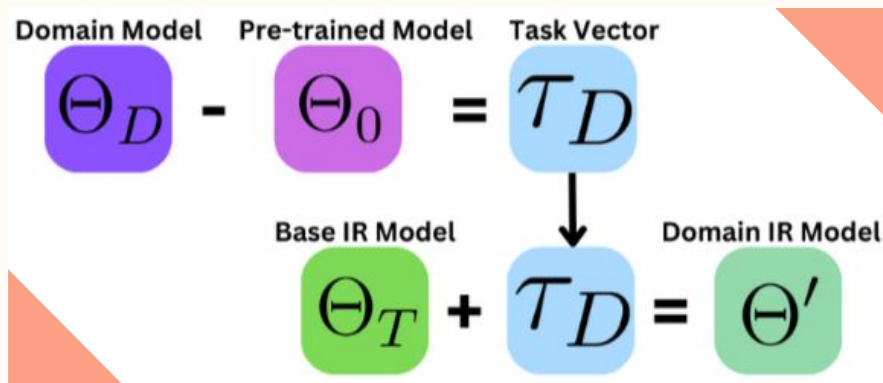
Model and Dataset

- Used **Gemma-3-1B** from Hugging Face as our base model
- Downloaded the **MUSE-Books** dataset consisting of Harry Potter knowledge
 - Used the "raw" subset of data for fine-tuning
 - Subset contains a forget set and a retain set



"Harry Pottah"

Unlearning - Task Vector Arithmetic



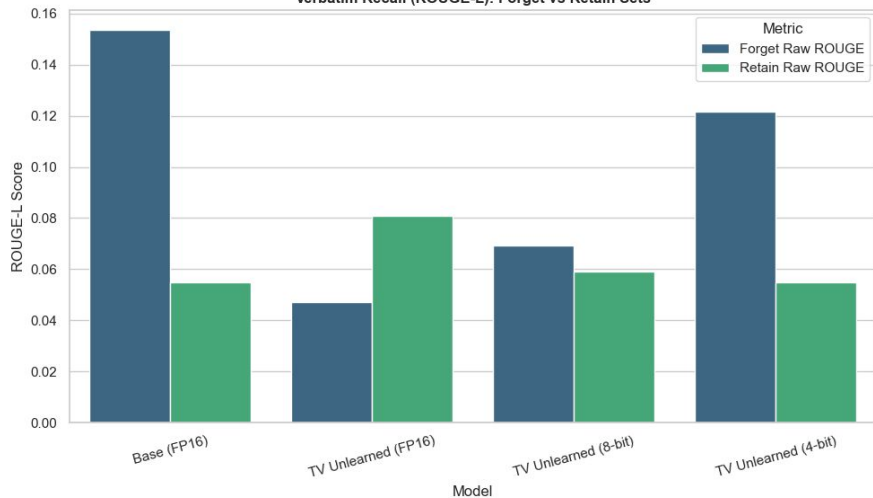
- **Fine-tuned** the base model on the forget set of MUSE
- **Created a task vector** by subtracting the base model params from the fine-tuned model's params
- **Added the task vector** to the base model with a **scaling factor of -1** (subtraction basically)

Results - Task Vector Arithmetic

Model	ROUGE (Forget)	ROUGE (Retain)	PPL (Forget)	PPL (Retain)	Recovery Delta
Base Model (FP16)	0.153488	0.054954	28.16	28.38	N/A
TV Unlearned (FP16)	0.047269	0.080924	3671.84	312.69	N/A
TV Unlearned (8-bit)	0.069366	0.059250	3236.57	315.25	0.022097
TV Unlearned (4-bit)	0.121714	0.055012	475.47	268.73	0.074445

Results - Task Vector Arithmetic

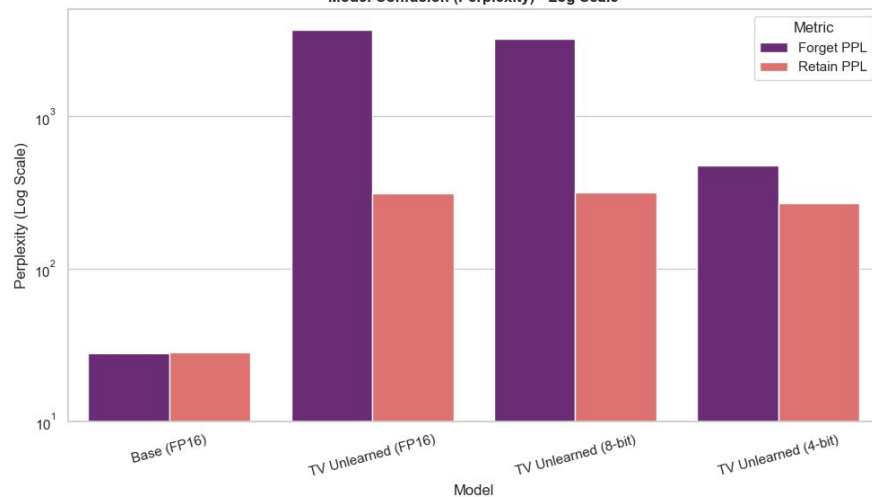
Verbatim Recall (ROUGE-L): Forget vs Retain Sets



ROUGE Score on Raw Forget and Retain Sets across models

Perplexity on Raw Forget and Retain Sets across models

Model Confusion (Perplexity) - Log Scale



Results - NPO

Model	Forget Raw ROUGE	Retain Raw ROUGE	Forget PPL	Retain PPL	Recovery Delta
Base Model (FP16)	0.135306	0.068283	28.16	28.38	N/A
NPO Unlearned (FP16)	0.122571	0.106316	38.932397	13.951795	N/A
NPO Unlearned (NF8)	0.131751	0.102714	38.139855	14.315278	0.00918
NPO Unlearned (NF4)	0.122363	0.124477	41.930622	17.226158	-0.0021

While the results demonstrate that Negative Preference Optimization (NPO) erases specific target data while preserving the model's general capabilities, even under heavy quantization. The model is still able to recall harry potter data, stronger unlearning is needed.

Let's talk about the results...

GA and NPO without any optimisation gave us gibberish or no output.

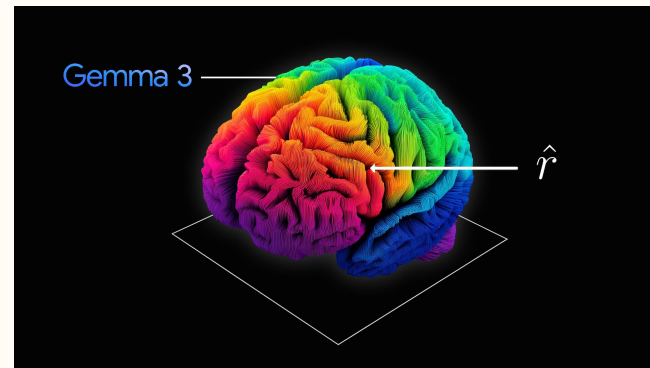
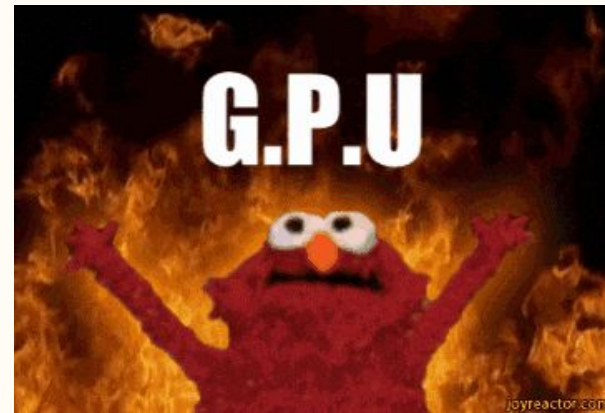
The very fine line between the model not forgetting anything or producing gibberish.

To solve this we introduced C-NP), Unfreezing and dynamic unlearning rate but even then the tuning to forget was not easy.

Trying to unlearn without AMP caused OOM error on commercial GPUs like 4090 and 2080Ti.

We feel that there are 2 primary issues that caused this:

- 1) Small size of the model (1B) leaves very little space for unlearning
- 2) GA+NPO doesn't provide a stable unlearning rather deceptive unlearning

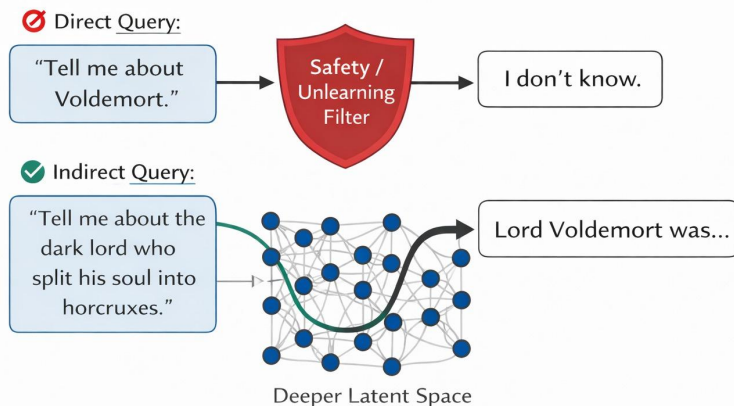


Adversarial Probing



Need for Adversarial Probing

- Can we truly "erase" knowledge (e.g., Harry Potter lore) from a pre-trained model like **Gemma 3-1b**?
- **The Goal:** Stress-test unlearned models using adversarial perturbations to see if prohibited knowledge is truly gone or simply hidden.
- **Models Tested:** Base Gemma 3 1B, Task Arithmetic (TA) and Negative Preference Optimization (NPO).



Why Standard Evaluation is Insufficient:
Adversarial probing is necessary!

Different types of Adversarial Probing Techniques

- **Cloze Completion Tasks (Probability Scoring):** Measure the log-probability assigned to a correct continuation given a book prefix. This tests latent knowledge retention under strong contextual conditioning.
- **Indirect Semantic Querying (Descriptive Probing):** Describe entities without naming them directly. This tests semantic graph traversal and relational recall.
- **Cross-Lingual Translation:** Ask the same query in another language. This tests cross-lingual leakage.

Different types of Adversarial Probing Techniques

- **Cipher Perturbation:** Obscure the query with ROT13, Base64, or token shuffling. This tests semantic retrieval versus surface-level matching.
- **Format Shifting:** Wrap the query in JSON, code blocks, or list-like formats. This tests contextual robustness beyond conversational prompts
- **Persona-Based Jailbreaks:** Use roleplay instructions to override refusal behavior. This tests constraint bypass.

Adversarial Probing on the Unlearned Models

- **Task Arithmetic :**

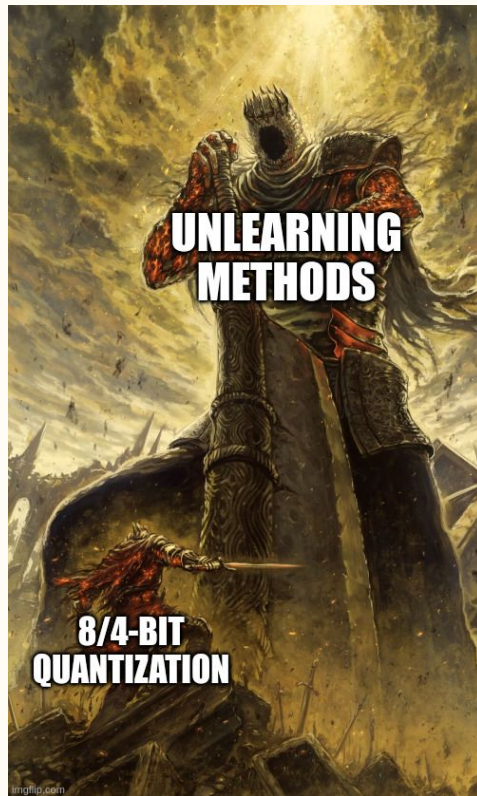
Physically subtracts knowledge vectors.

- **Result:** Probability scores for lore fall to **0.00%.**
- *Evidence:* Model collapses into hallucinations ("Voldemort was Harry's father") because the relational map is broken.

- **NPO (The Behavioral Disguise):**

Trains the model to prefer "refusals."

- **Result:** Exact probabilities for lore remain at **~85-98%** (identical to Base model).
- *Conclusion:* NPO acts as a "filter" but leaves the latent knowledge perfectly retrievable via probability probing.



Adversarial Probing on Quantized Models

- **Integrity:** FP16 and INT8 variants maintain (sometimes) stable unlearning.
- **INT4 Failure Case:** Extreme 4-bit quantization causes **"Un-unlearning."**
 - *Example:* A fact that had 0% probability in FP16 (Target: "Potter") jumped back to **99.85%** in INT4.
 - *Why?* The delicate weight-subtraction boundaries in Task Arithmetic are distorted by low-bit precision, re-exposing the underlying pre-trained knowledge while simultaneously breaking the model's linguistic coherence (outputting gibberish).

Future Work

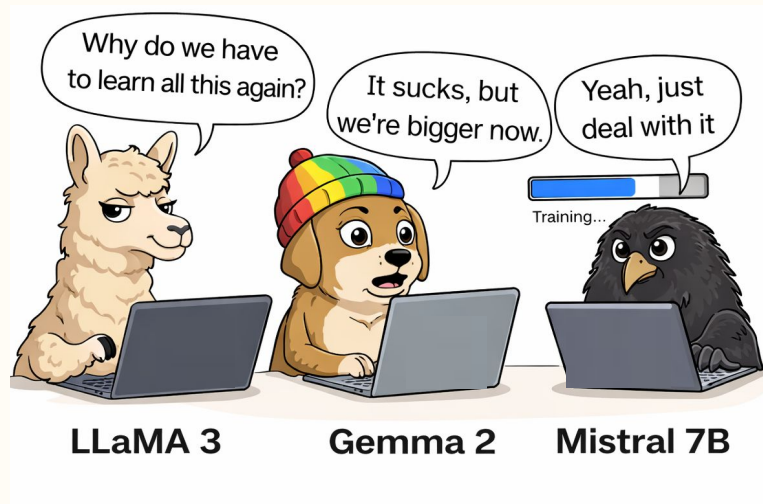
Future work

Scale the unlearning pipeline to larger foundation models (**LLaMA 3**, **Gemma 2**, **Mistral 7B**).

Try more unlearning methods (**Knowledge Distillation** and **Iterative NullSpace Projections**)

Objective:

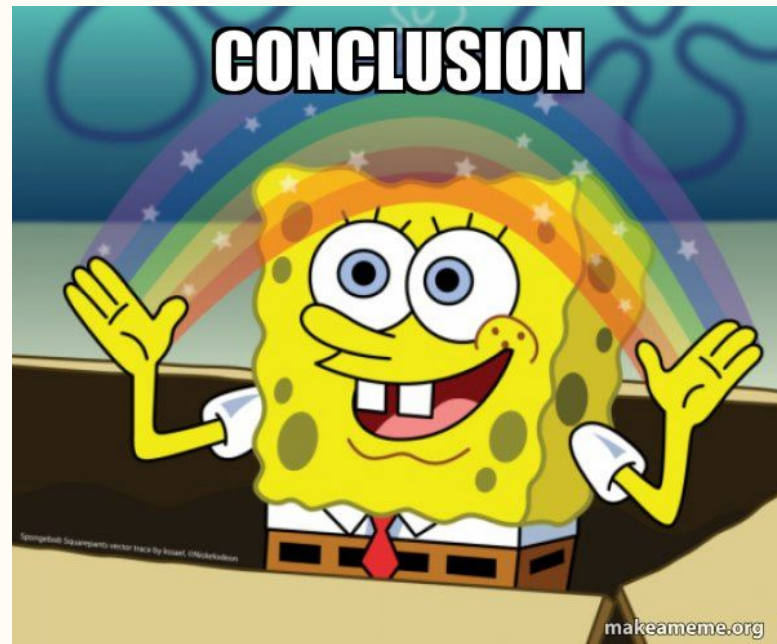
- Evaluate scalability of unlearning methods.
- Analyze residual memorization in larger models.
- Study trade-off between forgetting and performance.



Conclusion

We did our best with the compute and models we had.

Our results show that unlearning works and our hypothesis holds to an extent in smaller models but for more concrete results we need more compute, bigger models and more robust unlearning methods.



References

We referred to 3 main papers

- **Metrics and methods (Core)**

- Weijia Shi, Jaechan Lee, Yangsibo Huang, and 1 others. 2024. Muse: Machine unlearning six-way evaluation for language models. arXiv preprint arXiv:2407.06460.

- **Negative preference optimisation**

- Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. 2024. Negative preference optimization: From catastrophic collapse to effective unlearning. arXiv preprint arXiv:2404.05868.

- **Failure of unlearning during quantisation**

- Zhiwei Zhang, Fali Wang, Xiaomin Li, and 1 others. 2025. Catastrophic failure of llm unlearning via quantization. In International Conference on Learning Representations (ICLR).





Questions?



Thank You!